Work by Speech

User Guide

Work by Speech is the first program in the world that allows efficient work on a computer by speech without needing a keyboard and mouse.

1. **Work by Speech requirements:**
   - Windows 7 SP1, 8.1, 10 or 11
   - .NET Framework 4.7.2 or higher
   - Any microphone

2. **Recommended microphone for speech recognition**
   - This application works well even with the cheapest microphones. The main disadvantage of using such microphones is the fact that they have low sensitivity and SNR (signal to noise ratio). It means that in order to achieve high speech recognition accuracy, a cheap microphone must be placed close to the mouth.
   - A good microphone has sensitivity around -40 dB while having high SNR. Such a microphone works well even if placed approximately 20 inch away from the user. A microphone with a sensitivity rating of -40 dB is more sensitive than -50 dB, and -50 dB is more sensitive than -60 dB.
   - When it comes to microphone pickup patterns, it’s best to choose a cardioid microphone for speech recognition, because it’s most sensitive to sound coming from the front.
   - If you are in a noisy environment:
     - Use a headset microphone or a cardioid desktop microphone placed close to your mouth.
     - A microphone with high sensitivity isn’t necessary.
     - You should speak into the microphone with a normal sound volume.

3. **Setting up microphone:**

   1) Make sure that your microphone is connected to your computer.

   2) Work by Speech uses the default microphone for speech recognition. Go to Start -> Settings -> System -> Sound and make sure that your microphone is set as the default input device.

   3) If your microphone has a knob that allows changing volume, set it to the highest volume possible.

   4) Make sure that your microphone is working correctly by going to Start -> Settings -> System -> Sound. Under **Input**, select your microphone. Speak into the microphone and watch the **Input volume** indicator. If the blue bar moves, Windows is receiving audio from the microphone. For a more thorough test, click **Start test**, speak for a few seconds, then click **Stop test**. Windows will show the percentage of sound it detected.

4. Modes

- Work by Speech works in 3 modes:
  -  **OFF** – represented by red microphone icon (also in the system tray). The only voice command available in this mode is *Start speech recognition*.

  -  **Command** – represented by green microphone icon. This mode allows efficient work on a computer by using voice commands.
  -  **Dictation** – represented by blue microphone icon. This mode allows dictation.

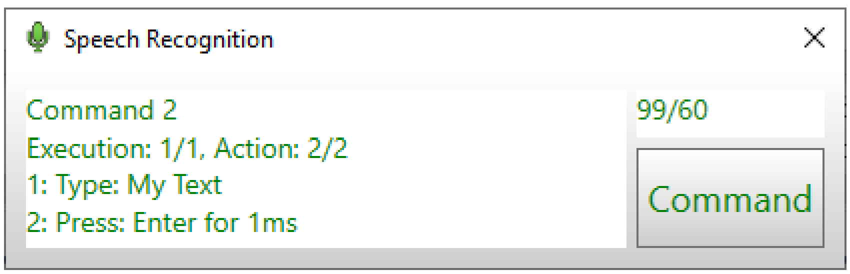
5. Speech recognition window

- It shows recognized speech, recognition confidence, required recognition confidence and current mode name in the *Mode* button.
- If required recognition confidence is not fulfilled, recognized speech appears as red text and action is not performed.
- If required recognition confidence is fulfilled, recognized speech appears as green text and action is performed.
- You can change the mode by using voice command, by clicking the *Mode* button or by right clicking Work by Speech icon in the system tray.
- You can hide this window by clicking X in window's top right corner or by saying *Hide speech recognition*.

6. Custom commands

- Custom commands are voice commands that you can create by yourself to perform actions that you want.
- These commands automate and speed up repetitive tasks.
- Custom commands are stored in profiles.
- Each profile can be associated with any program or a specific application. If a profile is associated with a specific program, all commands within it work only in that program.
- If more than one profile is associated with the same program, all of them will be used.
- Custom commands have higher priority than built-in commands. If a custom command is named the same as a built-in command, only the custom command is executed when its name is spoken.

- You may enable, disable and duplicate a profile by right clicking it and choosing the desired option.
- You may enable, disable, copy and paste a command by right clicking it and choosing the desired option.
- To execute a custom command, simply say its name while in *Command* mode.
- You may execute a command more than once if its max executions number is set to a value higher than 1.
- Do not set max executions number to a value higher than necessary.
- To increase the command's execution number say *twice* or *x times* after saying the command name.
- Multiple executions examples for command named *My Command* with max executions set to 50:
    - *My Command* twice – executes *My Command* twice
    - *My Command* 8 times – executes *My Command* 8 times
    - *My Command* 50 times – executes *My Command* 50 times
- The speech recognition window allows monitoring of custom commands execution.



7. Macro Recorder

- You can use the Macro Recorder to record a sequence of mouse movements, mouse clicks and keystrokes. The Macro Recording window can be opened by clicking the *Record* button in the Custom Command Management window.
- The Macro Recorder accurately records mouse movements when *Record Mouse Movements* option is selected. Mouse clicks are always recorded with cursor coordinates, so selecting this option is not recommended for recording simple mouse actions.
- Recorded macros can be added to a custom voice command and edited later. Other types of actions can be added to it. You can also share your macros with friends by using the *Export Profiles* feature.

8. Mousegrid

- The mousegrid divides the screen into up to 2550 figures.
- Each figure contains up to 2 characters. To read them you need to know the mousegrid alphabet which is described in the *Mousegrid Alphabet* document. Reading a string inside a figure moves the mouse cursor to the center of that string and performs mouse action that was chosen earlier by a voice command.
- You can move the mousegrid by saying up, down, left, right.

- Mousegrid types are:

    - **Hexagonal** – consists of regular hexagons (hexagons may be almost regular, because their height may be changed so screen is equally covered by them – the same thing can happen to the figures in other mousegrid types)
    - **Square** – consists of squares.
    - **Square horizontal precision** – consists of squares that are positioned in a way that increases mouse horizontal precision.
    - **Square vertical precision** – consists of squares that are positioned in a way that increases mouse vertical precision.
    - **Square combined precision** – consists of squares that are positioned in a way that increases mouse horizontal and vertical precision.
- Mousegrid lines can be used by every mousegrid type, but not hexagonal.
- Font color, font size and background color can be changed.
- Changing font size changes background surface area as well.
- The lower the desired figures number, the larger the font size should be for Mousegrid to work properly when square horizontal, vertical or combined precision is used. For example, if the desired figures number is 1500, the font size should be set to 16. If the desired figures number is 500, the font size should be set to 24.

9. Smart mousegrid

- When Smart mousegrid is enabled, mouse control is sped up based on how you use each application. The more often a figure is chosen, the shorter code word(s) it has.
- Smart mousegrid works separately for each application.
- Smart mousegrid data is collected separately for each mousegrid type and desired figures number.

10. Open [application name] command

- This command can open any application that has a shortcut file (.lnk) in *C:\ProgramData\Microsoft\Windows\Start Menu\Programs*.
- If you can't open a program by using this command, you need to copy a shortcut to this application to *C:\ProgramData\Microsoft\Windows\Start Menu\Programs*.
- You need to restart Work by Speech after installing a new program to be able to open it with "Open [application name]" command .

11. Using Work by Speech by multiple users on the same computer

- Restart Work by Speech whenever a different user starts using the program.
  If you suddenly switch to a different speaker without restarting the program, the decoder may be biased by the previous speech, temporarily reducing speech regonition accuracy.